01 · LESSON

Gas exchange in leaves

9B Plant growth · Lesson 3 of 7

Date _____ Class _____

Today I can

- ☐ I can describe how stomata and guard cells control gas exchange in a leaf.
- ☐ I can explain how temperature and concentration difference affect diffusion, and why a plant loses more water on a hot day.

What happens today

1. Do Now: a leaf under the microscope
2. Learn: Stomata and guard cells · Diffusion · and more
3. Practical: Looking at stomata
4. Challenge, then the exit ticket

Student Book pages 25 · Workbook pages 18

What I learned today (one or two sentences)

02 · KEY VOCABULARY

Key Vocabulary

Term	Definition
stomata	tiny holes in a leaf that let gases in and out (one is a stoma)
guard cell	one of the pair of cells that opens and closes a stoma
diffusion	the overall movement of particles from where they are more concentrated to where they are less concentrated
concentration	how many particles of a substance there are in a certain space
gas exchange	the swapping of gases between a leaf and the air
water vapour	water in the form of a gas
evaporation	a liquid turning into a gas
wilt	to droop because the cells have too little water
epidermis	the outer layer of cells of a leaf

03 · QUICK REVIEW

Quick Review

a leaf under the microscope

1 What is this hole in the leaf surface called?

2 Name two gases that pass through it.

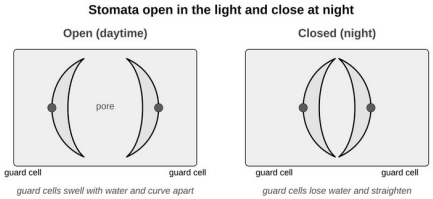
3 Predict when the hole is shut: day or night?

04 · CORE

Core 1 of 5

Everyone does these. Answer on the card or under it.

1 The diagram shows a stoma in the lower epidermis of a leaf at two different times of day.



a) State the function of the guard cells.

Core 2 of 5

Everyone does these. Answer on the card or under it.

1 *continued*

b) Name two gases that are exchanged between the leaf and the air through stomata.

c) Which diagram shows a stoma at night? Give a reason for your answer.

Core 3 of 5

Everyone does these. Answer on the card or under it.

2 Complete the sentences. Use the word bank: concentration, fall, photosynthesis, rapidly, respiration, rise, slowly.

In bright light, _____ takes up carbon dioxide from the air inside the leaf.

This causes the _____ of carbon dioxide inside the leaf to _____.

So carbon dioxide diffuses more _____ into the leaf from the air.

Core 4 of 5

Everyone does these. Answer on the card or under it.

3 Tick one box in each row to show whether the statement is true or false.

Statement about diffusion	True	False
Particles move from where they are less concentrated to where they are more concentrated.		
Diffusion is faster at higher temperatures.		
A bigger difference in concentration makes diffusion faster.		
Once particles are evenly spread they stop moving.		
Leaves are thin so gases do not have to diffuse far.		

Core 5 of 5

Everyone does these. Answer on the card or under it.

4 Practical: stomata on the lower surface of a leaf.

a) Draw one stoma with its two guard cells. Label the guard cells and the pore.

Draw here.

b) Record the number of stomata you can see in the field of view.

Count 1	Count 2	Count 3	Mean

Challenge

When Core is finished. Full sentences.

C1 Most leaves have far more stomata on their lower surface than on their upper surface. Suggest two reasons why this helps the plant.

C2 A student counted a mean of 32 stomata in a field of view with an area of 0.5 mm². The lower surface of the leaf had an area of 40 cm² (4000 mm²). Estimate the number of stomata on the lower surface of the leaf. Show your working.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What is the job of the guard cells?

2 Name two gases that are exchanged between a leaf and the air through its stomata.

3 Explain why a leaf loses more water on a hot day than on a cool day.

Today I can: Goal 1 ☐ Goal 2 ☐